

An Adaptive Security Selection Framework for Resource-Constrained IoT Devices Using the LinUCB Algorithm

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Abstract—In this paper, an adaptive cryptographic selection is investigated to navigate the security-efficiency trade-off in resource-constrained IoT devices. An operational scenario is considered wherein an edge device experiences continuous network threat spikes alongside depletion of battery life. To address this challenge, the cipher selection process is modelled as a contextual multi-armed bandit problem. A hardware-efficient LinUCB policy is designed by employing Sherman-Morrison matrix updates to ensure real-time processor feasibility. At each transmission round, the system evaluates residual battery life and environmental threat levels to dynamically select an optimal cipher from a set of lightweight and standard algorithms. Evaluated against trace-driven simulations parameterized by benchmarked hardware energy profiles from prior literature, simulations reveal that the agent successfully learns a linear approximation of the physical constraints while avoiding software-inefficient ciphers and severe late-stage battery penalties. Furthermore, experimental results demonstrate that the proposed adaptive strategy outperforms static baselines and rigid heuristics. By dynamically shifting between lightweight and heavyweight encryption, the framework achieves a near-Oracle cumulative security while preventing premature battery exhaustion.

I. INTRODUCTION

The Internet of Things (IoT) devices are significantly constrained in energy, memory, and computational resources, while simultaneously demanding a high level of security. Cryptographic techniques are employed to secure communication in IoT devices, and the selection of a specific algorithm is largely influenced by the device's lifetime and computational costs [1], [2]. Lightweight cryptography emerged as a promising solution for IoT devices [1], [2]. However, the selection of an appropriate algorithm involves balancing the required security level with resource costs, and static approaches cannot offer an optimal trade-off for the devices [2], [3].

Prior work has extensively examined the relationship between cryptographic design and resource consumption in constrained systems. Hardware evaluations indicate that lightweight ciphers such as PRESENT require significantly less area and power compared to AES on resource-limited

platforms [1]. Similarly, studies in wireless sensor networks demonstrate that encryption choice strongly influences operational lifetime, with lightweight Feistel-based ciphers such as Piccolo and XTEA enabling longer device operation than heavier alternatives [4]. Other research has explored the broader security-energy trade-off, showing that strong authenticity guarantees and energy efficiency cannot always be simultaneously maximised [5]. Formal modelling approaches further reveal that varying security levels can produce substantial differences in network lifetime [3]. Traditional game-theoretic models have also been used to analyse attacker-defender interactions and collaborative detection mechanisms in IoT systems [6], [7].

Motivated by these trade-offs, several adaptive security frameworks have been proposed. Hybrid cryptographic architectures dynamically adjust encryption parameters based on device state and environmental context [8], while autonomic frameworks employ machine learning models such as decision trees to select cryptographic algorithms according to device conditions and application requirements [9]. Additional studies explore adaptive IoT security architectures and dynamic cryptographic selection mechanisms for constrained edge devices [10], [11]. However, most existing approaches rely on supervised learning or predefined rules that require prior training and assume relatively stable operating environments, limiting their applicability in highly dynamic IoT scenarios.

The multi-armed bandit (MAB) framework provides a natural model for sequential decision-making under uncertainty [12]. In this framework, an agent repeatedly selects an action and observes a stochastic outcome while balancing exploration and exploitation to maximise cumulative reward [12]. Bandit algorithms are particularly suitable for IoT environments because they require minimal computational overhead, operate without labelled data, and can adapt online to changing system conditions [12], [13]. Consequently, they have been applied to a wide range of IoT and edge computing problems, including budgeted resource allocation [14], decentralised task

offloading [15], and context-aware network service selection [16]. Bandit frameworks have also been explored in security applications such as adaptive intrusion detection [17]–[20], and recent work extends contextual bandits to non-stationary environments, where reward distributions evolve [13]. Despite these advances, surveys of IoT security architectures indicate that many cryptographic solutions remain largely static and rarely incorporate online learning methods capable of continuous adaptation during deployment [2].

This paper addresses this gap by modelling the problem of cryptographic algorithm selection for IoT devices in the decision-making process. In this paper, a contextual multi-armed bandit framework has been proposed to select cryptographic algorithms based on device states and environmental contexts, without any prior training or rule-based methods.

The main contributions of this paper are threefold. First, the problem of adaptive cryptographic algorithm selection in IoT devices is formulated as a contextual multi-armed bandit problem, enabling online, autonomous optimisation of the security–efficiency trade-off. Second, a hardware-efficient linear decision framework is developed that efficiently handles complex, non-linear physical characteristics—such as exponential battery voltage drops and asymmetric threat risks—without requiring memory-intensive non-linear solvers. Third, the proposed approach is evaluated through trace-driven simulations parameterized by empirical edge microcontroller benchmarks from published literature under a continuous, non-stationary threat model. The results demonstrate that the LinUCB agent significantly outperforms both static deployments and rigid, rule-based heuristics, successfully avoiding premature device failure and achieving near-optimal, oracle-level performance along the Pareto frontier.

The remainder of this paper is organised as follows. Section II formulates the adaptive cryptographic selection problem and the contextual bandit model. Section III presents the proposed hardware-efficient LinUCB algorithm and its theoretical decision boundaries. Section IV details the experimental simulation setup and cryptographic baselines. Section V presents the results and discusses the security–energy trade-off. Finally, Section VI concludes the paper.

II. PROBLEM FORMULATION

A. System Model

As illustrated in Fig. 1, the framework comprises an IoT Device Layer for state extraction and cryptographic execution, and a Learning & Decision Layer for online algorithm selection. At transmission round $t \in \{1, \dots, T\}$, the device samples an operating context vector $x_t \in \mathcal{X} \subset \mathbb{R}^d$:

$$x_t = [B_t, \tau_t, 1.0]^\top \in \mathbb{R}^3, \quad (1)$$

where $B_t \in [0, 1]$ represents the normalized residual battery level, $\tau_t \in [0, 1]$ denotes the estimated ambient threat level, and 1.0 provides an affine bias term.

Before data transmission, the device selects a cryptographic cipher a_t from candidate action set $\mathcal{A} = \{a_1, a_2, \dots, a_K\}$. Each cipher $k \in \mathcal{A}$ has static benchmarked parameters derived

from hardware profiles (Table I): baseline energy consumption E_k , execution latency L_k , and intrinsic security rating S_k .

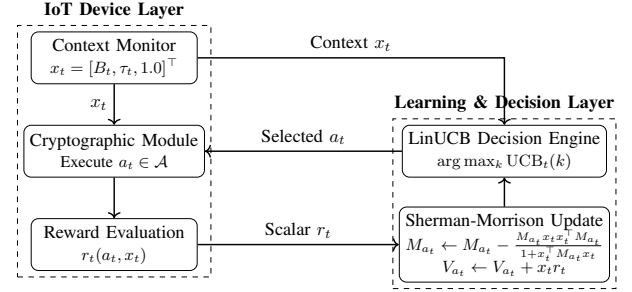


Fig. 1. System architecture and runtime operational workflow of the adaptive cryptographic algorithm selection framework.

B. Objective and Resource Constraints

The selection policy π must balance security effectiveness against energy and latency penalties. Weighting coefficients $\omega_E = 0.3$ and $\omega_L = 0.1$ normalize physical units (μJ and ms) relative to security metric $S_k \in [0, 12]$, preventing single metrics from dominating the objective landscape. Let C_π and M_π denote runtime computational complexity and memory footprint. The policy optimizes cumulative expected utility under device constraints $C_{\max}$ and $M_{\max}$:

$$\begin{aligned} \max_{\pi} \mathbb{E} \left[\sum_{t=1}^T r_t(a_t, x_t) \right] \\ \text{s.t. } C_\pi \leq C_{\max}, \quad M_\pi \leq M_{\max}. \end{aligned} \quad (2)$$

While the learning policy optimizes a linear model, the physical evaluation environment applies dynamic non-linear penalties, including quadratic threat utility and exponential battery depletion near capacity limits.

III. PROPOSED ADAPTIVE SELECTION FRAMEWORK

A. Contextual Bandit Formulation

Each cipher $k \in \mathcal{A}$ corresponds to an arm in a contextual multi-armed bandit. Upon selecting arm a_t at round t , the system observes a scalar reward:

$$r_t = S_{a_t} \tau_t^2 - \Omega(B_t) \omega_E E_{a_t} - \omega_L L_{a_t}, \quad (3)$$

where $\Omega(B_t) = \exp(2.5(1.0 - B_t))$ acts as an exponential battery penalty factor that prioritizes energy conservation as $B_t \rightarrow 0$. The agent assumes a linear model for each arm k :

$$\mathbb{E}[r_t \mid x_t, a_t = k] = x_t^\top \theta_k, \quad (4)$$

where $\theta_k \in \mathbb{R}^d$ is an unknown regression parameter vector estimated online.

B. LinUCB with Sherman-Morrison Inversion

Standard ridge regression requires an inverse covariance computation costing $\mathcal{O}(d^3)$ per step, which exceeds $C_{\max}$ on constrained microcontrollers. To eliminate matrix inversion, the Sherman-Morrison rank-1 update is used to update the

inverse covariance matrix $M_k(t) = A_k(t)^{-1} \in \mathbb{R}^{d \times d}$ in $\mathcal{O}(d^2)$ time.

For each candidate cipher k , the weight vector estimate is $\hat{\theta}_k(t) = M_k(t)V_k(t)$, where $V_k(t) \in \mathbb{R}^d$ accumulates context-reward products. The upper confidence bound score is computed as:

$$\text{UCB}_t(k) = x_t^\top \hat{\theta}_k(t) + \alpha \sqrt{x_t^\top M_k(t) x_t}, \quad (5)$$

$$a_t = \arg \max_{k \in \mathcal{A}} \text{UCB}_t(k), \quad (6)$$

where $\alpha > 0$ controls the exploration-exploitation parameter.

Following execution of a_t and observation of reward r_t , model state variables are updated:

$$M_{a_t}(t+1) = M_{a_t}(t) - \frac{M_{a_t}(t)x_t x_t^\top M_{a_t}(t)}{1 + x_t^\top M_{a_t}(t)x_t}, \quad (7)$$

$$V_{a_t}(t+1) = V_{a_t}(t) + x_t r_t. \quad (8)$$

C. Theoretical Complexity and Decision Thresholds

The algorithm requires $\mathcal{O}(Kd^2)$ arithmetic operations per decision step and maintains $\mathcal{O}(Kd^2 + Kd)$ scalar values in SRAM. For $K = 4$ candidate ciphers and context dimension $d = 3$, this consumes fewer than 50 floating-point variables, strictly satisfying $C_{\max}$ and $M_{\max}$ constraints.

For any two candidate ciphers i and j where $S_i > S_j$ and $E_i > E_j$, the policy switches preference to the higher-security cipher i when:

$$\frac{\tau_t^2}{\Omega(B_t)} > \frac{\omega_E(E_i - E_j) + \omega_L(L_i - L_j)}{S_i - S_j}. \quad (9)$$

The LinUCB engine learns this operational boundary adaptively without explicit offline parameter calibration.

Algorithm 1 LinUCB with Sherman-Morrison Updates

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1: Initialize:  $M_k = \frac{1}{\lambda} I_d$ ,  $V_k = \mathbf{0}_d$  for all  $k \in \mathcal{A}$ 
2: for  $t = 1, \dots, T$  do
3:   Observe context vector  $x_t = [B_t, \tau_t, 1.0]^\top$ 
4:   for each algorithm  $k \in \mathcal{A}$  do
5:      $\hat{\theta}_k = M_k V_k$ 
6:      $\text{UCB}_t(k) = x_t^\top \hat{\theta}_k + \alpha \sqrt{x_t^\top M_k x_t}$ 
7:   end for
8:   Select cipher  $a_t = \arg \max_{k \in \mathcal{A}} \text{UCB}_t(k)$ 
9:   Execute encryption  $a_t$  and observe reward  $r_t$ 
10:   $M_{a_t} \leftarrow M_{a_t} - \frac{M_{a_t} x_t x_t^\top M_{a_t}}{1 + x_t^\top M_{a_t} x_t}$ 
11:   $V_{a_t} \leftarrow V_{a_t} + x_t r_t$ 
12: end for

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IV. EXPERIMENTAL SETUP

The proposed framework is evaluated through the simulation of a resource-constrained IoT device operating under changing threat conditions. The goal is to analyse whether contextual bandit learning can improve the trade-off between security strength and energy consumption compared to static cryptographic policies.

A. Simulation Environment

The simulated device operates with a battery capacity of 7,000 energy units for a maximum of $T = 5000$ transmission rounds. This is scaled relative to the operational costs in Table I, which imposes a strict hardware constraint on the device and ensures a sufficient horizon for observing the impacts of selection policies.

Results are averaged across 30 trials to ensure statistical stability. Multi-objective weighting parameters are set to $\omega_E = 0.3$ and $\omega_L = 0.1$, and the LinUCB exploration parameter of $\alpha = 2.0$.

B. Cryptographic Algorithms

A candidate set $\mathcal{A} = \{\text{AES}, \text{PRESENT}, \text{Piccolo}, \text{XTEA}\}$ is selected to represent three distinct classes of ciphers: standard software-efficient (AES), ultra-lightweight software-oriented (Piccolo, XTEA), and lightweight hardware-oriented (PRESENT).

To accurately model the constraints, the parameters in Table I are derived from empirical hardware benchmarks on edge microcontrollers [1], [2], [4]. Capturing these software execution realities is critical. For example, PRESENT is highly efficient on dedicated hardware (ASICs); its bit-level permutations incur severe energy and latency penalties when executed on other general-purpose microprocessors [4]. In contrast, the byte-oriented operations of AES execute much more efficiently in software, while Piccolo and XTEA provide minimal resource footprints in exchange for reduced security margins.

TABLE I
BASELINE CRYPTOGRAPHIC ALGORITHM CHARACTERISTICS EVALUATED ON EDGE MICROCONTROLLERS.

Algorithm	Energy (e_k)	Latency (l_k)	Security (s_k)
AES	3.5	3.0	12.0
PRESENT	26.5	5.0	9.5
Piccolo	1.0	1.0	7.0
XTEA	3.8	3.2	7.0

The intrinsic security ratings $S_k \in \{12.0, 9.5, 7.0, 7.0\}$ in Table I correspond to NIST standard key-length security levels and algebraic cryptanalysis margins reported in [1], [2], establishing a normalized dimensionless scale across standard and lightweight ciphers.

C. Context Representation and Threat Model

To formulate the contextual bandit problem, the context vector at each round t is created to encapsulate the critical real-time state of both the device and the environment, such that:

$$x_t = [b_t, \tau_t, 1]^\top, \quad (10)$$

where $b_t \in (0, 1]$ represents the normalised residual battery level, and $\tau_t \in (0, 1]$ is the estimated environmental threat level, and the constant 1 acts as a bias feature. Having b_t

allows the agent to learn on its own the penalty of using high-energy algorithms.

To rigorously evaluate the algorithm's adaptability, the threat model avoids unrealistic deterministic step-functions. Instead, the environmental threat τ_t is modelled as a continuous, noisy, non-stationary process. This is achieved by superimposing three realistic network dynamics:

- 1) **Baseline Traffic:** A sinusoidal wave representing cyclic daily network activity, defined as $\tau_{base}(t) = \mu_{base} + A_{base} \sin(2\pi t/T_{cycle})$.
- 2) **Active Exploits:** Structured adversarial attacks are modelled using Gaussian distributions to simulate the smooth onset, peak, and decay of an active intrusion phase, defined as $\tau_{attack}(t) = \sum A_i \exp\left(-\frac{(t-\mu_i)^2}{2\sigma_i^2}\right)$.
- 3) **Sensor Noise:** To simulate the uncertainty of real-world Intrusion Detection Systems (IDS), additive Gaussian noise $\epsilon_t \sim \mathcal{N}(0, \sigma_{noise}^2)$ is applied at every transmission round.

The final bounded threat context observed by the agent is given by $\tau_t = \max(\min(\tau_{base}(t) + \tau_{attack}(t) + \epsilon_t, 1.0), 0.01)$. This chaotic, continuous threat environment critically tests the bandit's ability to maintain low regret and avoid unnecessary high-energy cryptographic executions when faced with ambiguous or noisy situational awareness.

The choice of keeping dimensions of the context vector low ($d = 3$) is deliberate and made to satisfy the strict hardware constraints. As Sherman-Morrison matrix updates scale as $\mathcal{O}(d^2)$, restricting the feature space to the most essential variables makes sure that the agent doesn't cause too much computational latency. This way, the LinUCB agent acts as a first-order approximator, which can effectively navigate complex non-linear problems without severe memory and processing overhead, as seen in high-dimensional or deep learning architectures.

D. Reward Function

To properly evaluate the algorithm's performance, the ground-truth R_t^* is formulated as a highly non-linear function reflecting the physical realities of the environment. To recreate the voltage drop-off and depletion of the lithium-ion batteries, the dynamic resource penalty employs an exponential curve as in (11).

$$\Omega^*(b_t) = e^{\beta(1-b_t)} \quad (11)$$

Where β controls the steepness of the penalty threshold. Furthermore, to capture the risk of active network exploits compared to average background noise, the effective security must scale quadratically with the observed threat level as shown in (12).

$$S^*(a_t, \tau_t) = S(a_t) \cdot \tau_t^2 \quad (12)$$

The contextual bandit is then evaluated against the non-linear environment ground truth as follows in (13).

$$R_t^* = S^*(a_t, \tau_t) - \Omega^*(b_t)\omega_E E(a_t) - \omega_L L(a_t) \quad (13)$$

Most importantly, the LinUCB remains unaware of these dynamics. Its hypothesis space consists solely of the linear combination $x_t^\top \theta_a$ as established in Section III. This mismatch forces the bandit to empirically learn a lightweight first-order approximation of the given environment. As a result, it demonstrates the viability of the proposed architecture for edge hardware.

E. Adaptive Selection Strategy

Algorithm selection follows the LinUCB strategy defined in Section III. At each round, the agent computes the Upper Confidence Bound for each cipher using the incrementally updated inverse matrix $M_k(t)$, selecting the algorithm that maximises this bound as expressed in (14).

$$a_t = \arg \max_{k \in \mathcal{A}} \left(x_t^\top \hat{\theta}_k + \alpha \sqrt{x_t^\top M_k(t) x_t} \right) \quad (14)$$

A small sensitivity analysis on α with $\alpha \in \{0.5, 1.0, 2.0, 4.0\}$ establishes $\alpha = 2.0$ as the optimal threshold; lower values ($\alpha = 0.5$) will end up causing premature exploitation of sub-optimal arms during early threat spikes, while higher values ($\alpha = 4.0$) will incur excessive exploration penalties by repeatedly testing inefficient ciphers like PRESENT. Therefore, $\alpha = 2.0$ is used for all evaluations.

F. Baseline Policies

The LinUCB is compared against static baseline strategies (Static AES, PRESENT, Piccolo, and XTEA), representing conventional deployments where algorithms are always fixed regardless of conditions. In addition, the framework is benchmarked against an optimal Oracle policy, which possesses perfect environmental knowledge, and a rule-based Heuristic Threshold policy.

This Heuristic policy acts as a standard reactive Intrusion Detection System (IDS) toggle: *If $\tau_t > 0.5$, deploy AES; otherwise, deploy Piccolo.* Though this threshold is effective at reacting to threat spikes, it is blind to the device's physical battery state, establishing a much more realistic baseline for conventional security logic.

V. RESULTS AND DISCUSSION

A. Cumulative Reward Performance

Fig. 2 illustrates the cumulative reward obtained by each policy during the transmission phase. The reward obtained by the policies is a measure of the trade-offs between security, battery life, and computational latency. The LinUCB algorithm exhibits a reward trajectory that closely tracks the Oracle policy. When the device experiences active threat spikes, it adjusts its trajectory accordingly, ensuring high efficiency. Before extreme late-stage battery depletion ($t \leq 3000$), LinUCB tracks the Oracle's trajectory closely, achieving 97.19% of the Oracle's cumulative reward efficiency before the non-linear exponential penalty dominates near $B_t \rightarrow 0$.

On the other hand, the Heuristic Threshold algorithm deteriorates sharply after $t \approx 3000$ due to its inability to adapt to depleting battery levels under noisy threat observations.

Static AES, PRESENT, and XTEA suffer early collapse due to premature battery exhaustion. Piccolo achieves a stable trajectory because its low energy consumption minimizes the exponential battery penalty, though it does so at the cost of providing minimal security during major threat spikes.

To ensure the robustness of the exploration-exploitation trade-off, a sensitivity analysis was conducted on the upper confidence bound hyperparameter $\alpha \in \{0.5, 1.0, 2.0, 4.0\}$. The empirical results established $\alpha = 2.0$ as the optimal threshold; lower values ($\alpha = 0.5$) caused premature exploitation of sub-optimal arms during early threat spikes, while higher values ($\alpha = 4.0$) incurred excessive exploration penalties by repeatedly testing the highly inefficient PRESENT cipher. Consequently, $\alpha = 2.0$ was retained for all primary evaluations. Furthermore, sensitivity sweeps over energy weights $\omega_E \in [0.1, 0.5]$ demonstrated consistent policy behavior: increasing ω_E accelerates the transition to lightweight Piccolo under low battery, while lowering ω_E extends AES retention during intermediate threat periods without causing policy divergence.

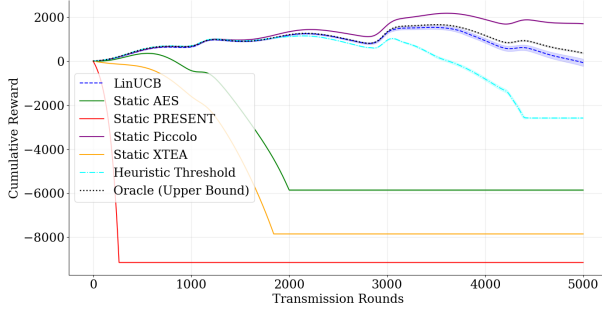


Fig. 2. Cumulative reward comparison of adaptive LinUCB and static cryptographic policies under dynamic threat conditions. Shaded error bands indicate ± 1 standard deviation across 30 independent Monte Carlo trials.

B. Battery Consumption Behavior

Fig. 3 tracks the residual battery life of each policy across the operational horizon. As configured from the benchmarks in Table I, static PRESENT, AES, and XTEA rapidly deplete the device’s battery, resulting in premature node failure before round $t = 2000$. The heuristic policy fails at $t \approx 4300$ because its threshold logic blindly deploys high-energy AES during late-stage threat spikes regardless of remaining energy. LinUCB survives the full $T = 5000$ horizon while closely matching the Oracle’s energy trajectory, extending operational lifetime significantly beyond all alternatives except ultra-lightweight Piccolo.

C. Adaptive Algorithm Selection

Fig. 4 illustrates the selection frequency of each cryptographic algorithm by the LinUCB policy across all 30 trials. Initial rounds exhibit active exploration across all arms. The agent rapidly learns to prioritize Piccolo during low-threat baseline periods to conserve energy, dynamically switching to

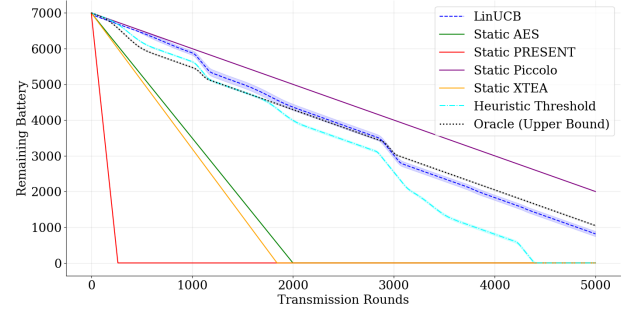


Fig. 3. Device battery trajectory under different cryptographic strategies. Shaded error bands indicate ± 1 standard deviation across 30 independent Monte Carlo trials.

AES during active threat spikes to maximize security margins. Because the bandit operates on continuous confidence bounds rather than rigid thresholds, it occasionally selects XTEA under intermediate context states. PRESENT is rejected after early exploration once the policy recognizes that its high software execution cost yields unfavorable reward trade-offs.

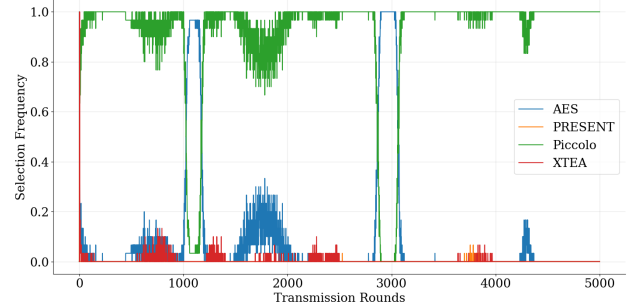


Fig. 4. Selection frequency of cryptographic algorithms under the LinUCB policy across 30 trials.

D. Security Achieved Over Time

Fig. 5 evaluates the cumulative security score achieved by each strategy. While static AES provides high instantaneous security, it depletes the battery early ($t \approx 2000$) and flatlines. Across the full $T = 5000$ transmission horizon, LinUCB achieves a cumulative security score matching 102.66% of the Oracle upper bound, confirming that the contextual policy successfully provides necessary security coverage during threat surges while preserving sufficient battery to remain operational throughout the entire horizon.

E. Security-Energy Trade-off

Fig. 6 presents the relationship between total energy consumed and total security achieved over the lifetime of the device, illustrating the Pareto frontier. Static baseline policies force a binary trade-off: sacrificing device lifetime for security (AES), or sacrificing security for longevity (Piccolo). LinUCB overcomes this trade-off by adapting to contextual

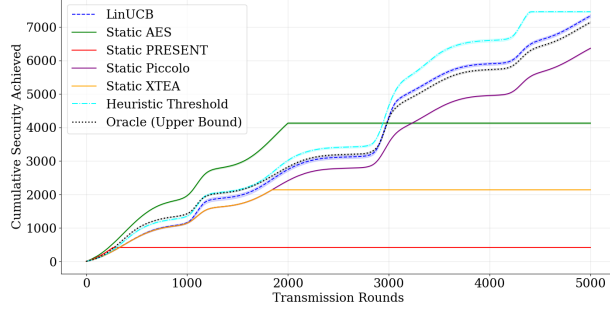


Fig. 5. Cumulative security achieved by each cryptographic strategy. Shaded error bands indicate ± 1 standard deviation across 30 independent Monte Carlo trials.

state transitions, tracking near-Oracle Pareto efficiency and achieving strong cumulative security with substantially lower total energy expenditure than static or rule-based methods.

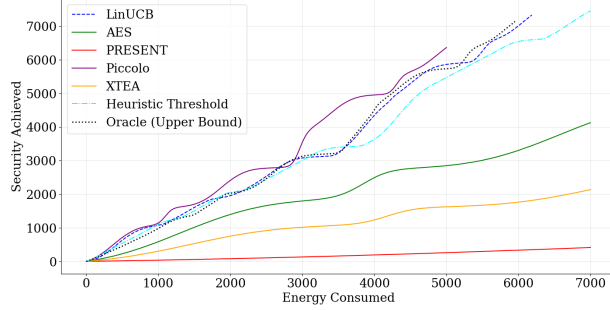


Fig. 6. Security versus energy trade-off for adaptive and static cryptographic strategies across 30 trials.

VI. CONCLUSIONS AND FUTURE DIRECTIONS

In this paper, a contextual multi-armed bandit framework was proposed to optimize the security-efficiency trade-off on resource-constrained IoT devices. Trace-driven simulations demonstrated that the proposed LinUCB policy with Sherman-Morrison rank-1 updates learns a lightweight linear approximation of complex non-linear battery and threat dynamics. The policy achieves 97.19% of the Oracle's cumulative reward efficiency prior to depletion and reaches 102.66% of Oracle cumulative security while guaranteeing node survival across the full 5000-round horizon.

Promising directions for future research include validating these dynamics on physical microcontroller testbeds with hardware power analyzers, extending the framework to decentralized multi-device networks, and evaluating adaptive policies under energy-harvesting constraints.

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